

MULTICORE BECOMES MAINSTREAM

Initially reserved for supercomputing and other specialised processing requirements, multicore has today become a mainstream technology for embedded applications. It provides a single consolidated platform for parallel computing, delivering a high throughput with power management that most embedded systems demand today



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Multiple processors or multiple cores are not new to the embedded space. Historically, these were used to deliver higher performance and higher reliability in complex embedded applications. Take the example of a very old technology of space shuttles in which five dedicated processors were used to run the shuttle functions.

Today's multicore architecture binds multiple processors on a single piece of silicon which not only delivers better performance per watt but also makes the whole thing look like a single machine to the application.

"A multicore architecture allows the processing to be distributed over multiple cores, thereby dividing up the workload to increase the throughput and reduce the required space and overall power consumption," explains Rohit Panikkar, technical marketing manager, National Instruments.

Mark Thronsdon, director of product marketing, MIPS Technologies, says, "The growing adoption of multicore solutions in a variety of embedded products is in response to rapidly increasing performance requirements. These requirements are driven by such trends as Web connectivity in a variety of traditional consumer electronics products, with

migration to 'cache-coherent' multicore platforms (also known as symmetric multiprocessing or SMP) as part of the solution."

"The increased processing performance and power efficiency of multicore processors are key factors driving their adoption in embedded systems," says Vasudevan Aghoramoorthy, vice president, semiconductor and system solutions, Wipro Technologies. According to analysts at independent research body Venture Development Corp. (VDC) in Massachusetts, revenues from multicore microprocessors in embedded applications in 2011 are projected to reach more than six times 2007 multicore revenues and more than 44 times 2006 level. They predict embedded CPU revenues to grow from \$50.7 million in 2006 to \$2.23 billion in 2011.

Multicore instead of single-core

When requirements push towards top-end performance, it becomes more and more costly and difficult to achieve performance gains with a single CPU core. Venkatesh Kumaran, country manager, Wind River, says, "As multicore chips provide better performance per watt of power consumption than single-core chips, these bring more CPU performance in the same size, weight and power budget than ever before."

All the tasks done using a multicore can be performed using a single-